Shared Intelligence, Competing Decisions: Policy Adoption and Congestion under Common Advice

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Abstract

A common decision policy may improve the accuracy of opportunity selection while concentrating participation in congestible opportunities. This paper studies a finite population choosing between a private-signal policy and a more accurate common-signal policy before signals are realized. An accounting identity separates expected intrinsic opportunity value from pairwise collision costs. In a binary-opportunity model, an interior symmetric mixed adoption equilibrium lowers expected participant payoff relative to the private-policy baseline whenever private signals are informative and the common policy is more accurate. The conclusion concerns commitment to executable policies, not unrestricted optimization after observing signals. We characterize pure adoption equilibria and fixed-count allocation, showing that the adverse effect need not be an artifact of independent adoption lotteries. A partial-commonality extension distinguishes comparisons at fixed adoption from comparisons with endogenous adoption. Exact enumeration, rational-arithmetic checks, and Monte Carlo calculations verify the analytical expressions. In an illustrative calibration, equilibrium adoption is approximately 40.5 percent, compared with a payoff-maximizing independent adoption rate of 15.3 percent. These calculations are synthetic and do not estimate losses from deployed artificial intelligence. The results clarify how accuracy, dependence, and strategic flexibility jointly determine the economic value of shared advice.

Keywords: algorithmic monoculture; congestion games; information externalities; policy adoption; common signals.
JEL classification: C72, D62, D83.

1 Introduction

Common analytical systems increasingly motivate a question that cannot be answered by measuring prediction accuracy alone: what happens when many decision makers act on similar recommendations? An opportunity can be attractive before a population responds and less attractive after that response. Shared evidence, similar decision rules, and scarce capacity can therefore interact even when the underlying predictions are informative.

This paper develops a deliberately small model of that interaction. Participants choose one of two executable policies before uncertainty resolves. One follows an independent private signal; the other follows a more accurate common signal. Both policies target one of two opportunities, and a participant's payoff declines linearly in the number of other participants choosing the same opportunity. The object of interest is the adoption game over these policies. Adoption is voluntary, but execution is committed: participants cannot combine signals or override the selected policy after observing its recommendation.

The main result is a closed-form characterization of symmetric adoption. When its equilibrium is interior, expected participant payoff is strictly below the outcome in which everyone

uses the private policy. The mechanism is a congestion externality: an adopter values its own improvement in selection quality but does not internalize the costs of its increasingly correlated action for others. This result is not a claim that more information is intrinsically harmful. An unrestricted planner could ignore a new optional signal, and a richer agent could implement a different mapping from information to action.

Three distinctions are central. First, an equilibrium of policy selection differs from an equilibrium of downstream actions. Second, common model use differs from coordinated allocation: a single provider could recommend different actions to different users. Third, participant payoff is not automatically economy-wide welfare. Throughout, welfare language refers to aggregate participant payoff and requires real resource costs, rather than transfers, for a social-surplus interpretation.

The paper supplies a transparent analytical example, finite-population extensions, and reproducible numerical verification. It does not assert priority for the general claim that shared algorithms can lower welfare, which is established in the literature. The analysis isolates assumptions under which a particular adoption externality occurs and identifies comparisons for which a simple monotonic claim about commonality fails.

2 Related literature

The traffic paradox of Braess et al. (2005) establishes that expanding feasible routes can worsen equilibrium travel times. Acemoglu et al. (2018) study informational versions in which drivers learn about available routes. Our policy-menu expansion is analogous at the level of equilibrium deterioration but is not a road-topology result or a theorem about route knowledge.

The closest conceptual precedent is Kleinberg and Raghavan (2021), who show that adoption of a common ranking algorithm can reduce collective hiring performance even when that algorithm is individually more accurate. The present binary-signal example belongs to the same family of externalities. It uses explicit pairwise collisions to separate intrinsic selection quality from congestion and to compare mixed adoption, fixed adopter counts, and partial commonality.

The value of public information depends on strategic interaction and the relevant efficiency benchmark. Angeletos and Pavan (2007) study information use in economies with complementarities or substitutes. Morris and Shin (2002) analyze public information with coordination incentives; Svensson (2006) cautions against a general anti-transparency interpretation. Coordination and congestion have different incentive structures, so the signs of comparative statics cannot be transferred without checking the model.

Strategic flexibility is especially important. Baek et al. (2026) show how congestion-aware differentiation can improve hiring outcomes relative to naive common targeting. Their analysis motivates the explicit commitment restriction here. Perdomo et al. (2020) formalize performative prediction, in which deployment changes the distribution being predicted. Our model is static, but the distinction between pre-response accuracy and post-response payoff is consistent with that broader concern.

There are also direct applications. Zhang et al. (2024) examine shared prediction displays in congestion games. Englander et al. (2025) estimate the value of public and private information in a congested fishery; their results do not support a universal negative value of public information. Chiang et al. (2026) study recommendation-induced congestion in school matching. A recent working paper by Chi and Yang (2026) addresses shared intelligence and firm entry; its abstract describes precision benefits, dependence costs, and endogenous adoption. Only its indexed abstract was accessible in preparing this version, so theorem-level overlap has not been assessed. Hedden and Raghavan (2026) provide a broader critique of unconditional objections to monoculture. These precedents make a narrowly stated contribution and an explicit novelty audit necessary.

3 Environment and accounting

3.1 Payoffs and collisions

Let $N \geq 2$ participants choose from K opportunities. Participant i selects $a_i \in \{1, \ldots, K\}$, and $n_j = \sum_i \mathbf{1}\{a_i = j\}$ denotes occupancy. Opportunity j has intrinsic value v_j, possibly stochastic. With a constant per-competitor cost $c > 0$, payoff is

$$u_i = v_{a_i} - c(n_{a_i} - 1). \tag{1}$$

Define aggregate payoff $W = \sum_i u_i$, mean expected intrinsic value $\mu = N^{-1} \sum_i \mathbb{E}[v_{a_i}]$, and average pairwise collision probability

$$q = \frac{1}{N(N-1)} \sum_{i \neq j} \mathbb{P}(a_i = a_j). \tag{2}$$

No symmetry or independence is needed for these definitions.

Proposition 3.1 (Collision decomposition)**.** *For any joint distribution of values and actions satisfying* (1)*,*

$$\frac{\mathbb{E}[W]}{N} = \mu - c(N-1)q. \tag{3}$$

Holding N and c fixed, an architectural change reduces expected average payoff if and only if $\Delta\mu < c(N-1)\Delta q$.

Proof. Summing (1) gives $W = \sum_i v_{a_i} - c \sum_j n_j(n_j - 1)$. The second sum counts ordered pairs of equal actions, so its expectation is $N(N-1)q$. Division by N yields the result. □

The proposition is an accounting identity, not a causal-identification claim. In applications, estimating c requires a model of actual resource costs or a credible intervention; a correlation between crowding and payoff is insufficient because valuable opportunities attract participation.

3.2 Binary signals and policy commitment

There are two opportunities and a binary state θ, with equal prior probabilities, indicating which opportunity is good. The good opportunity has value $v_0 + d$ and the other value v_0, where $d > 0$. The policy menu contains H and A.

Assumption 3.2 (Information and timing)**.** The private policy H selects the good opportunity with probability a, independently across participants conditional on θ. The common policy A selects it with probability b, and all adopters of A receive the same correctness realization. The common realization is independent of all private realizations conditional on θ. Parameters satisfy $1/2 < a < b < 1$. Participants are risk neutral, know the model, and choose policies simultaneously before observing the state or any signal. The selected policy then executes its recommendation without an override. Policy adoption is free.

This is a restriction on executable decision procedures. The menu does not include signal pooling, congestion-aware recommendations, or action revision. Consequently the equilibrium statements below do not characterize unrestricted information acquisition or post-signal Bayesian action choice. They can describe delegated policies selected before a decision window closes, but whether such commitment is realistic is an empirical question.

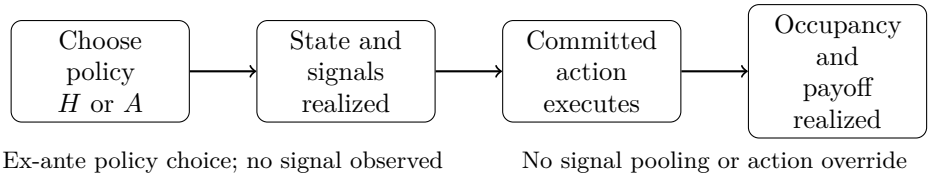


Figure 1: Timing of the policy-commitment game. The restriction on downstream execution defines the scope of the equilibrium theorem.

4 Symmetric adoption and efficiency

Let $p \in [0,1]$ be the common independent probability of choosing A, and let $w(p) = \mathbb{E}[W]/N$. Define

$$q_{HH} = a^2 + (1-a)^2, \qquad q_{HA} = ab + (1-a)(1-b), \qquad q_{AA} = 1, \tag{4}$$

$$L = c(N-1), \qquad \delta = d(b-a), \tag{5}$$

$$\alpha = q_{HA} - q_{HH} = (b-a)(2a-1), \qquad \beta = 1 - 2q_{HA} + q_{HH}. \tag{6}$$

Both α and β are positive. For β, writing $t = 2a-1$ and $s = 2b-1$ gives $\beta = (1+t^2-2ts)/2 > (1-t)^2/2 > 0$. Pairwise conditioning and Proposition 3.1 give

$$q(p) = (1-p)^2 q_{HH} + 2p(1-p)q_{HA} + p^2 = q_{HH} + 2\alpha p + \beta p^2, \tag{7}$$

$$w(p) = v_0 + da + \delta p - Lq(p). \tag{8}$$

Against opponents using adoption probability p, the expected gain from selecting A rather than H is

$$D(p) = \delta - L[\alpha + \beta p]. \tag{9}$$

Indeed, each opponent using H adds collision difference α, and each using A adds $1 - q_{HA} = \alpha + \beta$.

Theorem 4.1 (Symmetric equilibrium and expected loss)**.** *Under Assumption 3.2, the unique symmetric adoption probability in Nash equilibrium is*

$$p_E = \operatorname{clip}\left(\frac{\delta/L - \alpha}{\beta}, 0, 1\right), \tag{10}$$

where $\operatorname{clip}(x,0,1) = \min\{1, \max\{0,x\}\}$*. If*

$$L\alpha < \delta < L(\alpha + \beta), \tag{11}$$

the equilibrium is interior, and every participant has the same expected payoff satisfying

$$w(p_E) - w(0) = -L\alpha p_E < 0. \tag{12}$$

Proof. Equation (9) is strictly decreasing in p. An interior symmetric mixture requires indifference, $D(p) = 0$, which yields (10). If $D(0) \le 0$, $p = 0$ is the symmetric equilibrium; if $D(1) \ge 0$, $p = 1$ is. These statements include the endpoint indifference cases and establish uniqueness among symmetric profiles. At an interior equilibrium, $\delta = L(\alpha + \beta p_E)$. Subtracting $w(0)$ from (8) and substituting this equality yields

$$\delta p_E - L(2\alpha p_E + \beta p_E^2) = -L\alpha p_E.$$

Symmetry gives equal ex-ante payoffs. Strict negativity follows from $L, \alpha, p_E > 0$. □

The equilibrium is not claimed to be unique among asymmetric profiles. The theorem concerns ex-ante expectations, not a Pareto comparison in every realization. Moreover, a large loss under forced full adoption should not be substituted for the equilibrium loss when full adoption is not individually stable.

Proposition 4.2 (Restricted efficient adoption)**.** *The adoption probability maximizing average payoff among symmetric independent lotteries is*

$$p_S = \operatorname{clip}\left(\frac{\delta/L - 2\alpha}{2\beta}, 0, 1\right). \tag{13}$$

If both p_E and p_S are interior, $p_E - p_S = \delta/(2L\beta) > 0$.

Proof. From (8), $w'(p) = \delta - 2L(\alpha + \beta p)$ and $w''(p) = -2L\beta < 0$. The unique constrained maximizer is (13). Subtracting the two interior expressions proves the final statement. □

The qualifier "restricted" matters. A planner could coordinate adopter identities or actions rather than use independent lotteries. The next section treats the first of these alternatives.

5 Finite populations and pure adoption

Suppose exactly m named participants adopt A. Let $\bar{w}_m$ denote expected average payoff under this deterministic policy profile. Ordered-pair counting yields

$$\bar{w}_m = v_0 + da + \frac{\delta m}{N} - \frac{c}{N}\left[(N-m)(N-m-1)q_{HH} + 2m(N-m)q_{HA} + m(m-1)\right]. \tag{14}$$

Expanding (14) gives the useful expression

$$\bar{w}_m - w(0) = \frac{m}{N}\left[\delta - c\{2(N-1)\alpha + (m-1)\beta\}\right]. \tag{15}$$

Against exactly k adopting opponents, the private gain from adoption is

$$D_k = \delta - c\big[(N-1)\alpha + k\beta\big]. \tag{16}$$

Proposition 5.1 (Pure equilibrium counts)**.** *An interior adopter count $m \in \{1, \ldots, N-1\}$ supports a pure Nash equilibrium if and only if*

$$m - 1 \le T \le m, \qquad T = \frac{\delta/c - (N-1)\alpha}{\beta}. \tag{17}$$

The boundary count $m = 0$ is an equilibrium if $D_0 \le 0$, and $m = N$ is an equilibrium if $D_{N-1} \ge 0$. If $(N-1)\alpha > \beta$, every interior pure equilibrium has $\bar{w}_m < w(0)$.

Proof. A non-adopter faces m adopting opponents and does not want to adopt if $D_m \le 0$. An adopter faces $m-1$ adopting opponents and does not want to leave if $D_{m-1} \ge 0$. Substitution of (16) gives (17). The same logic proves the boundary conditions. At an interior pure equilibrium, $\delta \le c[(N-1)\alpha + m\beta]$. Substituting this bound into (15) gives

$$\bar{w}_m - w(0) \le \frac{cm}{N}\left[\beta - (N-1)\alpha\right] < 0.$$

□

For noninteger T in the interior range, the equilibrium count is $\lceil T \rceil$; an integer threshold can support two adjacent counts. The identities of adopters need not be unique. Proposition 5.1 supplies a sufficient condition under which the expected loss is not an artifact of symmetric mixed strategies.

Proposition 5.2 (Coordinating adopter counts)**.** *For an integer m and $p = m/N$,*

$$\bar{w}_m - w(p) = c\beta p(1-p) \geq 0. \tag{18}$$

A count maximizing (14) is obtained by checking the feasible integers adjacent to the clipped continuous maximizer

$$m_S^{\text{cont}} = \text{clip}\left(\frac{1}{2} + \frac{\delta/c - 2(N-1)\alpha}{2\beta}, 0, N\right). \tag{19}$$

Proof. Substitute $p = m/N$ into (8) and subtract from (15). The difference is $c\beta(m/N - m^2/N^2)$. Alternatively, independent adoption gives a binomial count M with variance $Np(1-p)$; the quadratic term in (15) implies a Jensen gap $c\beta \operatorname{Var}(M)/N$. Differentiating the concave quadratic in m gives the stated continuous maximizer, after which adjacent integers suffice. □

Count coordination is not necessarily incentive compatible. It is an allocation benchmark, not a decentralized equilibrium or an implementable mechanism without additional incentives.

6 Partial commonality and the role of adaptation

6.1 A controlled dependence comparison

Let $r \in [0,1]$. With probability r, all adopters share a single Bernoulli correctness draw with parameter b; otherwise each adopter receives an independent Bernoulli draw with the same parameter. The regime draw is independent of signals and adoption. Private signals remain conditionally independent. Marginal accuracy stays at b, while

$$q_{AA}(r) = b^2 + (1-b)^2 + 2rb(1-b), \tag{20}$$

$$\beta(r) = q_{AA}(r) - 2q_{HA} + q_{HH} = 2(b-a)^2 + 2rb(1-b) > 0. \tag{21}$$

All symmetric-adoption formulas continue to hold with $\beta(r)$, because they use only the pairwise collision probabilities. At fixed $p > 0$,

$$\frac{\partial w(p,r)}{\partial r} = -2Lp^2 b(1-b) < 0. \tag{22}$$

At an interior equilibrium, however,

$$\frac{\mathrm{d}p_E(r)}{\mathrm{d}r} = -\frac{(\delta/L - \alpha)\, 2b(1-b)}{\beta(r)^2} < 0. \tag{23}$$

Together with (12), this implies that equilibrium payoff increases with r on an interior branch: stronger dependence discourages enough adoption to reduce the equilibrium loss. This local comparative static should not be extrapolated across boundary regimes without using the clipped equilibrium expression. Fixed-adoption and endogenous-adoption comparisons answer different questions.

6.2 Execution outside the policy menu

If participants observe true opportunity values and can revise actions, they play a different congestion game. Let k choose the good opportunity. When both opportunities are occupied, the no-switch conditions are

$$c(2k - N - 1) \leq d \leq c(2k - N + 1). \tag{24}$$

The left inequality compares a good-opportunity user's current payoff with its payoff after switching; the right compares an other-opportunity user's payoff with entry into the good opportunity.

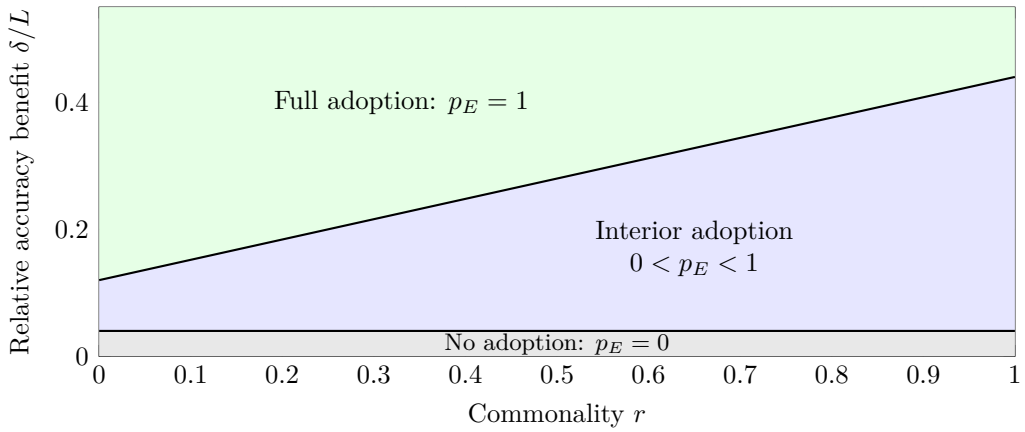


Figure 2: Equilibrium adoption regions with $a = 0.6$ and $b = 0.8$. The lower boundary is $\delta/L = \alpha = 0.04$ and the upper boundary is $\delta/L = \alpha + \beta(r) = 0.12 + 0.32r$. Boundary indifference is included in the clipped rule. The interior region satisfies the strict loss result.

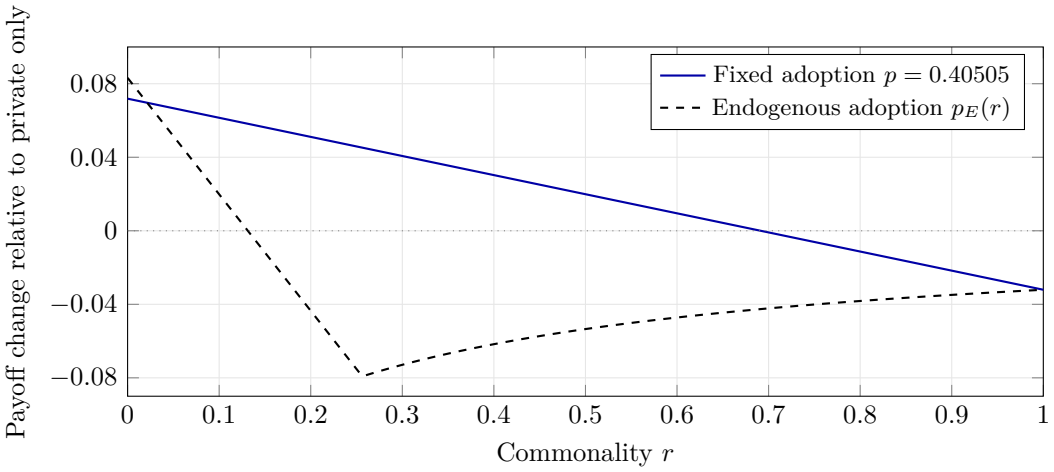


Figure 3: Comparative statics at fixed and endogenous adoption, holding marginal accuracies and the numerical calibration fixed. Greater commonality always lowers payoff at fixed adoption. On the interior-equilibrium branch it raises payoff by discouraging adoption. The kink occurs where full adoption ceases to be an equilibrium.

If $d \geq c(N-1)$, all participants selecting the good opportunity is an action equilibrium. A coordinated full-information planner instead maximizes

$$Nv_0 + dk - c\{k(k-1) + (N-k)(N-k-1)\}, \tag{25}$$

whose continuous optimum is $k = N/2 + d/(4c)$, clipped to $[0, N]$ and then checked at adjacent integers. Full information and efficient coordination are distinct. These benchmarks do not extend Theorem 4.1 to unrestricted agents; they demonstrate why the allowed policy set must be stated.

7 Computational verification

7.1 Design

The numerical example uses

$$N = 100, \quad a = 0.60, \quad b = 0.80, \quad v_0 = 9, \quad d = 2, \quad c = 0.02.$$

Table 1: Expected payoff and Monte Carlo verification. Payoffs are arbitrary model units per participant; each simulated condition uses 50,000 independent markets.

Regime	p	Exact	Monte Carlo	Standard error
Private only	0.0000	9.17040	9.17024	0.000271
Optimal independent adoption	0.1525	9.18883	9.18908	0.000434
Mixed adoption equilibrium	0.4051	9.13832	9.14127	0.001139
Full common adoption	1.0000	8.62000	8.62372	0.003565

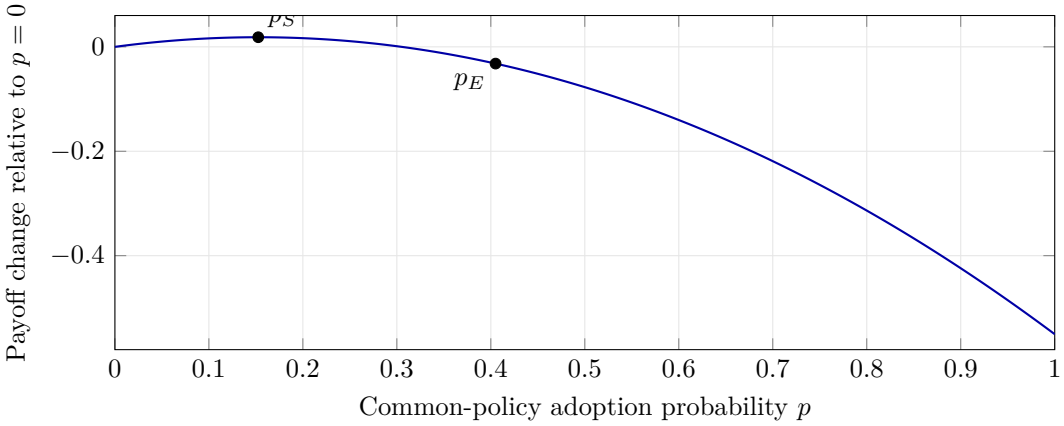


Figure 4: Exact payoff change from (8). The voluntary equilibrium lies beyond the adoption level maximizing expected payoff among independent symmetric lotteries.

Thus $L = 1.98$, $\alpha = 0.04$, and $\beta = 0.40$. The parameters illustrate an interior adoption equilibrium and a positive restricted efficient adoption rate; they are not estimated from a market. Python and NumPy generate 50,000 independent markets for each of four adoption probabilities, using generator seed 20261105 and batches of 1,000. Each market draws N independent adoption indicators, N private correctness indicators, and one common correctness indicator. The good opportunity can be assigned a fixed label without loss because the two state labels have symmetric payoffs.

Within each market, the realized mean payoff is computed from occupancy. Monte Carlo standard errors use the sample variance of market means divided by the number of markets. Individual outcomes within a market are dependent and are not treated as independent samples. The four conditions use successive unpaired draws of the same generator.

The implementation also performs exact enumeration for $N = 4$ at $p \in \{0, 0.17, 0.5, 1\}$, checks 6,564 interior cases on a parameter grid, and verifies 1,404 finite-count configurations with Python's exact rational arithmetic. The latter checks use three population sizes, three signal pairs, four cost values, and every feasible adopter count. These are numerical and algebraic consistency checks, not a proof-assistant certification or empirical validation.

7.2 Results

The symmetric equilibrium adoption probability is $p_E = 0.4050505$, whereas $p_S = 0.1525253$. The exact loss at equilibrium is 0.03208 per participant relative to the private-only baseline, approximately 0.35% of baseline payoff. Restricted efficient adoption gains 0.0184251 per participant. These percentages have no direct monetary or macroeconomic interpretation. Full adoption has a larger loss but is not an adoption equilibrium for this calibration.

The finite game has pure equilibria with 41 adopters, yielding mean payoff 9.138256, also below the private-only baseline 9.1704. The best fixed count is 16, yielding 9.189856. Thus the

qualitative loss is not created solely by the randomness of the symmetric mixture. The small advantage of fixed-count coordination over independent adoption is consistent with Proposition 5.2.

Exact enumeration differs from the analytical formula by at most 2.85×10^{-14} in floating-point computation. All four Monte Carlo discrepancies are below three standard errors. The mixed-equilibrium simulated mean is 2.59 standard errors above its expectation; the seed was retained rather than selected for closer agreement. Table 1 reports Monte Carlo uncertainty, not uncertainty about empirical parameters.

For a countercase, keep common-policy marginal accuracy at $b = 0.8$ but set $r = 0$. At full adoption, $q_{AA} = 0.68$, so expected payoff is 9.2536, exceeding the private-only baseline. In the fully informed action game, all participants choosing the good opportunity is an equilibrium with payoff 9.02 because $d = 2 > c(N-1) = 1.98$. A coordinated full-information allocation with 75 participants in the good opportunity yields 9.27. These comparisons involve different information and coordination constraints and should not be presented as estimates of one intervention.

7.3 Worked exact example

For the calibration, $a = 3/5$, $b = 4/5$, $c = 1/50$, and $d = 2$. Direct substitution gives

$$q_{HH} = \frac{13}{25}, \quad q_{HA} = \frac{14}{25}, \quad \alpha = \frac{1}{25}, \quad \beta = \frac{2}{5}, \quad L = \frac{99}{50}, \quad \delta = \frac{2}{5}.$$

The symmetric equilibrium and restricted optimum are therefore

$$p_E = \frac{401}{990}, \qquad p_S = \frac{151}{990}, \qquad w(p_E) - w(0) = -\frac{401}{12500} = -0.03208.$$

For pure adoption, $T = 401/10 = 40.1$. The only interior equilibrium count is consequently $m = 41$, since $40 \leq 40.1 \leq 41$. The sufficient loss condition holds because $(N-1)\alpha = 99/25 > 2/5 = \beta$. The fixed-count continuous optimum is 15.6; evaluating the adjacent integers selects $m = 16$. This example illustrates the proofs with exact fractions rather than relying on a simulation to establish the sign.

8 Discussion and limitations

The analysis identifies a conditional externality from adopting common executable advice. Better marginal signal accuracy does not determine the value of a population-wide deployment: dependence and congestion also matter. Nevertheless, commonality need not be harmful. With zero congestion, more accurate policies improve payoffs. At $a = 1/2$, the interior-equilibrium loss vanishes because $\alpha = 0$. If capacity scales with population so that $c = \kappa/(N-1)$, the factor L remains fixed and there is no mechanical population-size amplification.

The main modeling restriction is policy commitment. A user who can combine private and public information, predict congestion, or randomize execution has a larger strategy space. Theorem 4.1 does not characterize that game. A second restriction is homogeneous preferences and linear, symmetric congestion. Heterogeneous tasks and nonlinear costs alter the collision-to-payoff mapping. A third is the independence of private errors conditional on the state. Real human and algorithmic judgments can share errors even without a common provider.

The computational results are internal verification only. There are no live language-model calls, human participants, external datasets, or estimates of economy-wide losses. Nor does this static model demonstrate dynamic instability or chaos. A credible application would measure resource costs and manipulate evidence overlap, policy overlap, and congestion feedback separately. Group-level randomization is needed because decisions interact within markets; treating agents as independent observational units would understate uncertainty.

Publication-level novelty remains a separate question from mathematical validity. The broad mechanism is already established, and full-text comparison with the recent firm-entry work remains outstanding. The present contribution is a compact, explicit model with finite-population comparisons and a reproducible verification record. Its value as a journal contribution requires assessment against the closest literature, not merely successful compilation or numerical agreement.

9 Conclusion

A more accurate common policy can attract voluntary adoption while reducing expected participant payoff. In the restricted binary-signal game, the loss has a simple expression and survives at pure finite-population equilibria under a sufficient condition. Coordinating adopter counts improves outcomes relative to independent adoption with the same expected count. Partial-commonality analysis further shows that fixed-adoption and endogenous-adoption comparisons can point in opposite directions. These results support evaluating shared advice at the population level while retaining explicit assumptions about capacity, dependence, and the ability to revise actions.

Research transparency

Code and data. The manuscript source, analysis code, and synthetic results are available at https://github.com/triggeredcode/shared-intelligence-competing-decisions. No real-world observations or live-model experiments are reported.

AI assistance. AI was used in model development, analysis, coding, and manuscript preparation.

Status. This is a working paper and has not been peer reviewed.

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